Recursive Leverage in Yield-Bearing Stablecoins: Proof-of-Concept and Empirical Analysis

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1 Introduction

This project empirically studies how large holders use yield-bearing tokens (YBTs) to construct recursive looping strategies across Ethereum mainnet lending protocols. A YBT accrues yield passively (through staking, lending, or vault returns) while also serving as collateral. Some large holders exploit this dual property by depositing a YBT, borrowing against it, acquiring additional exposure, and repeating the process to increase effective carry at the cost of higher liquidation risk.

Looping refers to a recursive leverage strategy where the holder deposits a yield-bearing token as collateral, borrows a stablecoin against it, uses the borrowed stablecoin to purchase more of the same yield-bearing token, and re-deposits it as additional collateral, repeating the cycle to amplify effective carry at the cost of proportionally higher liquidation risk.

To preserve comparability across wallets and strategies, the study focuses exclusively on USD-denominated yield-bearing tokens on Ethereum mainnet, which is selected because the three candidate tokens originated there, their largest holder concentrations and longest transaction histories are on mainnet, and the Aave v3 and Morpho Blue mainnet deployments provide the deepest historical record for the lending activity under study. Three tokens are selected using the following criteria: top candidates by Ethereum mainnet circulating value and TVL; a minimum on-chain liquidity threshold; a minimum active holder count; active integration as collateral or borrowable asset on at least one major lending protocol; and activity throughout the sample period. Preliminary candidates are `sUSDe` (Ethena), `sUSDS`, and `syrupUSDC` (Maple Finance). The final token set will be confirmed at the

start of data collection.

The empirical analysis focuses on two major Ethereum mainnet lending markets where these tokens are actively used as collateral: Aave v3 and Morpho Blue. No additional protocols are included in order to keep the data pipeline tractable and ensure comparability across wallets.

2 Problem Statement

The primary contribution is an empirical characterisation of strategy archetypes among large YBT holders. A secondary contribution is testing whether on-chain risk indicators can predict short-horizon rebalancing and stressed risk states.

3 Research Questions

Using ML and own scripts, the following RQs will be answered:

- RQ1: What recurring leverage and looping strategy archetypes can be identified among large holders of major yield-bearing stablecoin tokens on Ethereum mainnet?
- RQ2: Which observable risk-management characteristics (leverage depth, collateral buffers, and rebalancing behaviour) differentiate these archetypes?
- RQ3: To what extent can on-chain state variables predict short-horizon rebalancing events and stressed risk states?

4 Methodology and Anticipated Outcomes

Our Master's Project will follow three main stages:

- Part I: Data Collection from Ethereum mainnet
  - **GraphQL**: protocol APIs for Aave v3 and Morpho Blue, providing structured borrow, supply, repay, and liquidation events at block-level resolution.
  - **Blockchain data providers** (e.g. Dune Analytics): SQL queries across protocol event tables for exploratory analysis and cross-protocol joins.
- Part II: Proof-of-Concept and Feature Engineering

  - Develop own script / bot for leveraging of YBT at major lending protocols (with and without flashloans)
  - Unit of observation: wallet–token–day. Raw on-chain events are transformed into a panel dataset with the following features: LTV ratio (time series), loop depth, distance to liquidation, rebalancing events, estimated carry spread, protocol concentration, known automation, and liquidation history.
- Part III: Machine Learning
  - Unsupervised clustering (HDBSCAN) to identify strategy archetypes.
  - Supervised classification (XGBoost) to predict stressed risk states, with SHAP for model interpretation.

4.1 Infrastructure Requirements

To deliver this project, we will require:

- Access to GraphQL, Dune Analytics.
- A Python environment with libraries for HDBSCAN, XGBoost, SHAP, and UMAP.
- Storage for the wallet–token–day panel dataset covering approximately 17 months of block-level event data.

4.2 Knowledge and Skills Requirements

Understanding of DeFi Protocols: Knowledge of lending protocol mechanics including collateral, LTV, liquidation thresholds, and health factor on Aave v3 and Morpho Blue.

On-Chain Data Engineering: Skills in querying Etherscan API, The Graph (GraphQL), and Dune Analytics (SQL) to reconstruct position histories at block-level resolution.

Machine Learning: Ability to apply and evaluate unsupervised clustering (HDBSCAN) and supervised classification (XGBoost) on structured tabular panel data, including handling class imbalance and interpreting models with SHAP.

4.3 Project Management and Tools

The project will be executed between Mai and September 2025, with bi-weekly meetings with the supervising assistant. We will use the collaboration tools: Overleaf, GitLab, Python (open-source pipeline).

4.4 Split of Responsibilities

- Morpho leveraged PoC Bot script - Dawid Rymarczyk
- Data collection and pipeline construction - Dawid Rymarczyk
- Feature engineering - Igli Kristo
- Unsupervised clustering and archetype analysis - Igli Kristo
- Supervised classifier training and evaluation - Dawid Rymrczyk
- Report writing - Dawid Rymarczyk and Igli Kristo

5 Timeline

- May 2025
  - Build Morpho leverage lending bot script (Proof of Concept) on YBT token flash loan.
  - Write scraper to collect data from the 3 largest yield-bearing stablecoins.
- June 2025
  - For each token, identify and select the 3–5 largest EOA whale accounts.
  - Begin data collection and feature engineering pipeline.
- July 2025
  - Complete feature engineering; construct wallet–token–day panel.
  - Train supervised learning models (e.g. XGBoost classifier) with SHAP explanations.
  - Run unsupervised learning models (e.g. HDBSCAN clustering, UMAP visualisation).
  - Potential feature engineering refinement

- August 2025
  - Analyse ML findings and interpret economic archetypes.
  - Adjust and refine the Proof of Concept bot based on insights from ML model analysis.
- September 2025
  - Finalise documentation and write-up of methodology and results.
  - Open-source code release and report submission.

6 Skeleton of Master Project

1. Introduction and Motivation
   - Why yield-bearing stablecoins and looping strategies emerge in DeFi.
   - Problem statement.
2. Background and Related Work
   - Overview of yield-bearing tokens,
   - Overview of lending protocols, and liquidation mechanics.
   - Literature on DeFi leverage, on-chain behavioural analysis, and ML on blockchain data.
3. Proof-of-concept
   - Leverage script with and without flashloan.
4. Methodology and Data Collection
   - Token selection criteria and final token set.
   - Wallet filtering methodology and sample of 20–30 strategic addresses.
   - Data sources and collection pipeline.
5. Feature Engineering
   - Construction of the wallet–token–day panel.
   - Definition and operationalisation of all eight features.

6. Unsupervised Clustering: Strategy Archetypes
    - HDBSCAN methodology and parameter choices.
    - Cluster validation: stability, economic interpretability, UMAP visualisation.
    - Empirical archetype taxonomy with quantitative risk profiles.
7. Supervised Classification: Stressed Risk State Prediction
    - Target variable definition and class imbalance handling.
    - XGBoost model training, temporal hold-out evaluation, and baseline comparison.
    - SHAP-based interpretation of predictions.
8. Discussion
    - Implications for DeFi risk monitoring and protocol design.
    - Limitations: sample size, loop depth approximation, data availability.
    - Directions for future work.
9. Conclusions
    - Summary of archetype findings and predictive model results.
    - Contribution to empirical DeFi research.